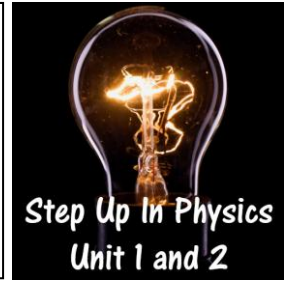


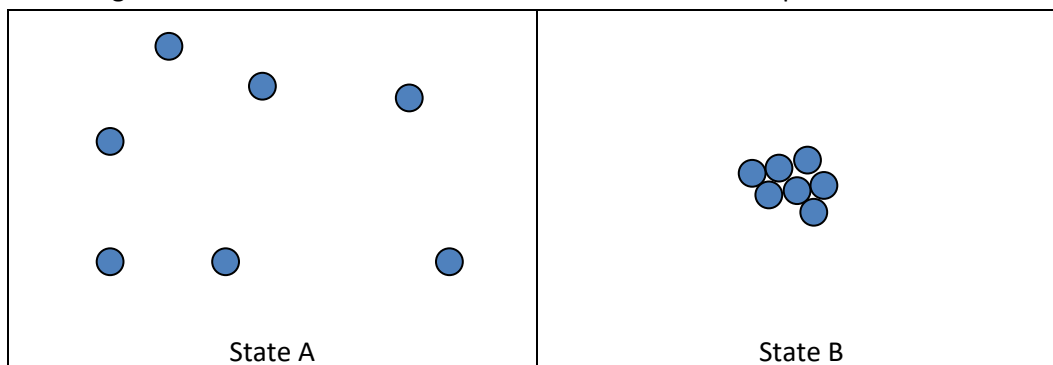
Kinetic Theory of Matter

Problems Worksheet



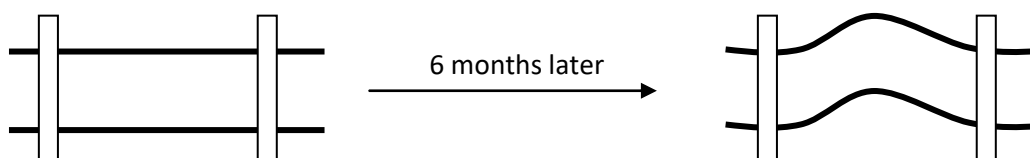
1. The kinetic theory of matter is a useful model for explaining and predicting the behaviours of the states of matter.
 - a. State one difference between the behaviour of particles in a solid compared to a liquid.
 - b. State one difference between the behaviour of particles in a liquid compared to a gas.
2. Heat from an oven will increase the temperature of a pizza while it cooks. Describe in terms of physics principles the difference between the temperature of the pizza and the heat used to cook the pizza.
3. The Celsius scale for measuring temperature was established by setting the zero point as the temperature that ice melts. The Kelvin scale zero point was based on a more elaborate measurement of average kinetic energy of particles that make up a substance. Describe why negative temperatures are allowed on the Celsius scale but not on the Kelvin scale.

4. The two diagrams below show the **same** substance at the **same** temperature but in different states.



- What is the most likely state of matter of State A? What is the most likely state of matter of State B?
- Which state (A, B or neither) has the highest kinetic energy? Justify your answer.
- Which state (A, B, or neither) has the highest potential energy? Justify your answer.
- Which state (A, B, or neither) has the highest internal energy? Justify your answer.

5. A sample of hydrogen gas in a cubic container had its temperature measured in all eight corners. The temperature was recorded as 87.0°C in each location.
- The velocity of one of the particles of the gas was measured and recorded as 350 ms^{-1} . What information (if any) does this tell you about the velocity of the other hydrogen atoms in the sample?
 - After a set time the temperature of the gas had reduced to 82.0°C . Describe the changes of the behaviour of the particles within the sample compared to when they were at a higher temperature.
6. A railway was installed in winter such that it was very snug, without any gaps between the metal rails. Six months later the rails had bent and deformed.



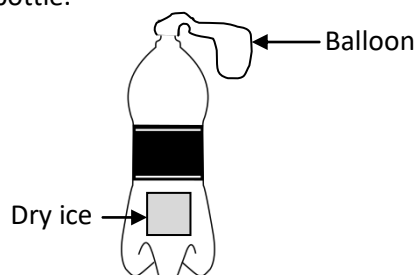
Explain why this occurred.

7. Over head power lines that are constructed in the summer months are always built with a little bit of slack when connecting the lines between poles. Explain why they are constructed this way.

8. A lower school student studying Geology performs an experiment designed to show how temperature changes can crack rocks. They place a glass rod in a Bunsen Burner flame and then plunge it into icy water. The glass rod cracks. Describe why this occurs.

9. Samantha tells Jack that the when the water rose in temperature from 20.0°C to 40.0°C that the average kinetic energy of the water doubled. Jack disagrees and says it only increased by 6.83 %. Who is right and why?

10. Carbon dioxide is unique in that it skips the liquid state of matter as it is heated from a solid to a gas (called subliming). A large chunk of solid carbon dioxide (dry ice) is placed in a small plastic bottle with a balloon secured over the open end of the bottle.



Describe the likely observation that would occur as the carbon dioxide sublimates. Explain your answer.